

Q1.

Carol needs access to her file for users who have age bigger than or equal to 21, whose department is electrical engineering, and who have completed EE 301. Carol can use attribute based encryption if the age condition is transformed from using a number and comparison operator to a valid attribute name like Age_21_Over. The other attributes would be EE_dept for the EE department condition, and EE301_Completion for the EE 301 condition. The access tree has an AND node at the root and has the three above attributes as leaves: Age_21_Over, EE_dept, and EE301_Completion.

Q2.

Attribute based encryption cannot enforce the full policy. The intern and file has already been accessed by the manager condition is dynamic and can be changed whenever a file is accessed by a manager. Attribute based encryption can only work and manage static and unchanging policies.

Q3. C

The data owner can go offline for ABE in cloud storage. The other options are all false.

Q4. A, C, D

Plaintext document, public parameters, and access policy are all required for ABE encryption.

Q5. C, D

To successfully decrypt a ciphertext, public parameters and an attribute set satisfying policy are required. Access policy is part of the system, but is not needed for decryption.

Q6. A, B, D

ORAM protects access frequency, access order, and read vs write types.

Q7. A, B, D

In Path-ORAM, the client must store encryption key, overflow blocks, and leaf/path assignment.

Q8. A, C

Server storage tree must have blocks that have equal size and dummy blocks may exist.

Q9. A, B, C

In data access, the client must download a full path, decrypt all downloaded blocks, and re-encrypt blocks.

Q10. A

Garbled circuits only hide inputs from the evaluator.

Q11. D

5 input wires + 4 internal wires + 2 output wires = 11 total wires

Each of the wires need 2 keys

$11 * 2 = 22$ total keys.

Q12. A, B, C, D

When garbling a gate, two keys per input wire, two keys per output wire, encrypt truth table rows, and randomly permute rows are all required.

Q13. A

The evaluator must receive one key per input wire

Q14. A, C

The final scheme requires an encryption scheme and a pseudorandom number generator.

Q15. B, C, D

The final scheme must have the keyword encrypted, the server must learn all positions where the keyword occurs, and client can decrypt the document.

Q16. A, D

For one out of two oblivious transfers, the sender must encrypt both messages and the receiver must obtain only one message.

Q17. A, D

Alice holds $a_1 = 0$, $b_1 = 1$ and Bob holds $a_2 = 1$, $b_2 = 1$. Let $c = a \times b$. What are possible output shares (c_1, c_2)

$$a = a_1 + a_2 = 0 + 1 = 1 \pmod{2} = 1$$

$$b = b_1 + b_2 = 1 + 1 = 2 \pmod{2} = 0$$

$$\text{So, } c = a \times b = 1 \times 0 = 0$$

Then calculate all possible pairs where $(c_1 + c_2) \pmod{2} = 0$

Possible pairs are $(0, 0)$ and $(1, 1)$ as they equate to 0 in c equation

Q18.

$$f(x) = 5 + 3x \text{ over } \mathbb{Z}_{11} \rightarrow f(x) = 5 + 3x \pmod{11} \text{ with } n = 3 \text{ and } t = 1$$

Shares at $x = 1, 2, 3$

$$f(1) = 5 + 3 = 8 \pmod{11} = 8$$

$$f(2) = 5 + 6 = 11 \pmod{11} = 0$$

$$f(3) = 5 + 9 = 14 \pmod{11} = 3$$

So shares are $(1, 8)$, $(2, 0)$, and $(3, 3)$

Reconstruct with two shares $(1, 8)$ and $(2, 0)$ with Lagrange.

$$s = 8 * (0 - 2)/(1-2) + 0 * (0 - 1)/(2-1)$$

$$s = 8 * 2 = 16 \% 11 = 5$$

The secret is therefore 5 and the secret has been successfully reconstructed.

Q19. A, C

Any $t + 1$ shares reconstruct, and utilization of a random polynomial are both true.

Q20. C

The polynomial degree is t

Q21.

Z7 with $a_1 = 2$, $a_2 = 3$, $b_1 = 4$, $b_2 = 1$

Reconstruction:

$$a = a_1 + a_2 = 2 + 3 = 5 \% 7 = 5$$

$$b = b_1 + b_2 = 4 + 1 = 5 \% 7 = 5$$

$$c = a + b = 5 + 5 = 10 \% 7 = 3$$

$$c_1 = a_1 + b_1 = 2 + 4 = 6 \% 7 = 6$$

$$c_2 = a_2 + b_2 = 3 + 1 = 4 \% 7 = 4$$

$$c_1 + c_2 = 6 + 4 = 10 \% 7 = 3$$

Shares of $c = a + b$: $c_1 = 6$, $c_2 = 4$

Reconstructed $c = 3$

Q22. A, C

Addition and scalar multiplication operations are local with no interaction.